

FOZ DO IGUAÇU, Brazil

WMO: 838270

Lat: 25.596S

Lon: 54.487W

Elev: 787

StdP: 14.28

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.8	42.5	33.5	28.9	46.5	37.0	33.3	48.2	20.3	75.3	18.3	73.3	3.4	190	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.5	95.4	73.3	93.5	73.4	91.7	73.5	78.9	88.8	78.0	87.8	77.1	86.7	9.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.9	139.3	83.7	75.2	136.3	83.0	74.0	130.6	81.6	43.0	89.1	42.0	88.1	41.0	86.8	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	15.8	13.7	DB	31.5	99.7	3.4	2.1	29.1	101.3	27.1	102.5	25.2	103.7	22.7	105.3	(4)
(5)			WB	31.0	81.1	3.4	1.4	28.6	82.1	26.6	82.9	24.7	83.6	22.2	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	79.5	78.9	77.0	72.4	65.5	63.4	62.8	66.5	69.0	73.3	75.2	78.0	(6)	
	DBStd	8.63	3.31	3.69	4.28	6.44	6.81	7.90	8.46	8.69	8.10	5.82	4.64	3.93	(7)	
	HDD50	21	0	0	0	0	1	5	10	4	1	0	0	0	(8)	
	HDD65	514	0	0	1	17	79	123	143	93	51	6	1	0	(9)	
	CDD50	7968	916	810	836	672	481	408	407	516	573	723	757	869	(10)	
	CDD65	2984	451	390	372	239	94	76	75	140	172	263	308	404	(11)	
	CDH74	25554	4208	3401	3018	1750	524	416	514	1244	1612	2292	2832	3743	(12)	
	CDH80	10621	1901	1499	1277	633	121	70	103	463	704	985	1211	1654	(13)	
(14) Wind	WSAvg	6.0	5.5	5.5	5.3	5.4	5.3	5.8	6.1	6.8	7.0	7.0	6.2	5.7	(14)	
(15) Precipitation	PrecAvg	69.6	6.5	5.6	4.9	5.4	6.4	5.2	4.1	3.5	6.0	8.5	6.3	7.3	(15)	
	PrecMax	100.5	14.5	11.8	12.9	12.9	15.9	12.1	9.4	13.4	12.8	23.8	13.4	18.0	(16)	
	PrecMin	49.1	0.7	0.4	1.7	0.9	0.0	0.4	1.1	0.6	1.6	1.7	1.2	0.9	(17)	
	PrecStd	12.8	2.9	2.6	3.0	3.0	4.4	2.8	2.1	3.2	2.8	4.3	3.3	4.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.9	98.5	96.4	92.8	88.0	84.5	85.9	91.6	96.8	97.0	96.5	96.8	(19)	
		MCWB	75.2	72.1	73.0	74.2	72.6	70.1	67.3	67.3	69.5	73.5	73.5	73.3	(20)	
	2%	DB	93.5	94.1	93.0	89.7	82.8	82.2	82.7	88.8	92.9	93.1	93.0	93.5	(21)	
		MCWB	75.3	74.6	73.3	74.1	70.2	68.4	66.6	66.8	68.4	72.5	72.3	74.3	(22)	
	5%	DB	91.6	91.3	89.8	86.9	79.8	79.1	80.3	85.7	88.1	89.6	89.8	91.3	(23)	
		MCWB	75.4	75.0	73.7	72.9	69.2	67.6	66.2	66.5	68.3	71.4	71.6	74.0	(24)	
	10%	DB	89.6	88.6	87.5	83.9	76.5	75.4	76.6	81.8	83.8	85.8	87.5	88.9	(25)	
		MCWB	75.0	74.4	73.2	72.2	68.0	67.1	64.8	65.7	67.2	70.3	71.2	73.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	79.7	78.9	77.8	75.5	73.1	71.1	72.3	74.9	77.4	78.2	79.6	(27)	
		MCDB	90.1	88.9	88.4	87.8	83.5	79.9	80.1	84.7	88.6	91.0	89.2	89.5	(28)	
	2%	WB	78.9	78.3	77.2	76.1	72.7	71.1	69.3	69.8	72.2	75.3	76.4	78.3	(29)	
		MCDB	88.9	87.5	87.2	85.5	79.6	78.4	78.3	82.2	83.7	87.9	86.8	88.1	(30)	
	5%	WB	77.8	77.2	76.0	74.8	70.8	69.3	67.5	68.1	70.4	73.6	74.7	77.1	(31)	
		MCDB	87.6	86.0	85.3	83.5	76.8	76.4	76.0	79.7	81.2	84.7	84.9	86.6	(32)	
	10%	WB	76.7	76.1	74.8	73.3	69.1	67.5	65.9	66.7	68.7	72.1	73.3	75.8	(33)	
		MCDB	85.9	84.4	83.1	80.8	74.7	73.5	74.1	77.8	78.9	81.5	82.6	84.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	19.2	19.6	19.1	17.7	17.7	20.0	22.5	22.0	20.3	21.2	20.2	(35)	
		MCDBR	22.8	23.1	23.1	21.8	20.7	20.7	23.1	26.4	27.8	26.2	25.7	24.2	(36)	
	5% WB	MCWBR	8.6	8.3	8.5	8.9	9.9	9.9	10.3	10.1	10.2	10.0	10.1	9.4	(37)	
		MCWBR	19.1	18.9	19.1	19.0	17.3	17.7	19.9	23.2	22.2	21.8	20.1	19.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.404	0.398	0.388	0.383	0.356	0.389	0.385	0.468	0.565	0.471	0.419	0.415	(40)	
		taud	2.430	2.445	2.464	2.448	2.493	2.398	2.386	2.108	1.859	2.190	2.341	2.378	(41)	
	Ebn at Noon		298	295	289	275	270	251	257	244	234	271	292	296	(42)	
		Edh at Noon	39	38	35	33	29	30	32	46	64	48	43	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		2043	1867	1706	1407	1045	905	1021	1259	1436	1643	2001	2049	(44)	
	RadStd		160	152	115	132	146	97	122	93	153	188	199	199	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page